

Shikhar Veeramachineni

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Education

B.Tech, Computer Science (AI & ML) — Vellore Institute of Technology Sept 2023 – Aug 2027
Currently pursuing a Bachelor's degree in Computer Science with specialization in Artificial Intelligence and Machine Learning. Coursework includes machine learning, deep learning, natural language processing, and computer vision.

Experience

Summer Trainee, Grant Thornton Bharat LLP June 2025 – July 2025

- Analyzed large-scale datasets related to PMAY (Pradhan Mantri Awas Yojana) beneficiaries and housing allocation records, supporting internal audits and government reporting through structured data analysis workflows.
- Built Python-based data processing pipelines using Pandas and NumPy to clean, transform, and aggregate raw datasets, enabling efficient analysis of beneficiary eligibility and allocation patterns.
- Developed data visualizations and analytical dashboards using Matplotlib to translate complex datasets into clear insights for consulting teams and decision-makers.
- Designed normalized SQL database schemas to store beneficiary and housing data, enabling efficient querying, aggregation, and standardized reporting across multiple project datasets.
- Collaborated with consulting and infrastructure teams to convert business requirements into data-driven insights, improving reporting accuracy and reducing manual analysis effort.

AI Intern, Coschool Apr 2023 – Jun 2023

- Evaluated generative AI systems for educational use cases by designing structured prompt engineering frameworks to test performance across summarization, tutoring, and knowledge generation tasks.
- Conducted systematic experiments to analyze model behavior under varying prompt structures, instruction styles, and edge-case inputs to measure robustness and response reliability.
- Benchmarked outputs from multiple large language models including ChatGPT and GitHub Copilot to assess response quality, reasoning consistency, and instructional alignment.
- Documented prompt engineering techniques and best practices to improve internal adoption of generative AI tools for content creation and educational workflows.
- Produced structured evaluation reports that helped teams understand model strengths, limitations, and potential integration strategies within educational platforms.

Projects

Emotio — Multimodal Emotion & Sentiment AI GitHub

- Built hybrid CV+NLP system integrating real-time facial emotion recognition with transformer-based sentiment classification for accurate mood detection and contextual user behavior analysis.
- Designed modular NLP pipelines including preprocessing, keyword extraction (KeyBERT), and structured sentiment scoring to improve interpretability, contextual relevance, and analytical consistency.
- Implemented optimized inference workflows using PyTorch and HuggingFace for scalable deployment across lightweight environments with reduced latency and efficient resource utilization.

IntelliPost — AI Document Intelligence System GitHub

- Developed an AI-powered document intelligence platform allowing authenticated users to upload PDFs and extract structured policy data through automated processing and organized content handling workflows.
- Built schema-based reasoning flows using Pydantic and Pydantic-AI to guide LLM summarization ensuring consistent outputs, validated inputs, and reliable content generation.
- Integrated FastAPI endpoints with asynchronous database storage to manage document access and navigation while maintaining smooth API responses and structured data persistence.

Traction AI — AI-Driven Revenue Operations Platform Github

- Designed backend workflows that convert a single AI prompt into investor-ready pitch decks and nine structured business documents with organized content generation.
- Built scalable document handling systems including secure APIs, user authentication, and automated slide export features for seamless content delivery.
- Designed AI system capable of generating 9 structured business documents from a single prompt, reducing manual business planning effort by 70 percent.

Automated Insurance Document Verification — OCR + NLP GitHub

- Designed an AI pipeline that extracts and validates insurance claim data using OCR-based document parsing and NLP-based entity extraction.
- Implemented rule-based and ML validation systems to detect inconsistencies in submitted claim documents.
- Built scalable document processing workflows capable of handling large batches of insurance forms with structured outputs.

Achievements

3rd Place — Post-a-thon Hackathon

Recognized for building IntelliPost, an AI-powered document intelligence platform demonstrating strong system design and practical AI deployment.

Technologies

AI/ML: PyTorch, TensorFlow, HuggingFace, Scikit-Learn, YOLO, OpenCV, FAISS

Backend: FastAPI, Pydantic, Pydantic-AI, LangGraph, Uvicorn, Groq API, Async APIs

Languages: Python, SQL, JavaScript, C++, Java

Domains: NLP, Computer Vision, AI Agents, RAG Systems, Model Deployment